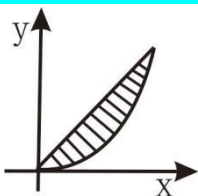


课时五 二重积分

练习 5.1: 计算二重积分 $\iint_D (x-1)d\sigma$ 区域 D 由 $y=x^2$ 和 $y=x$ 所围成的第一象限部分

解:



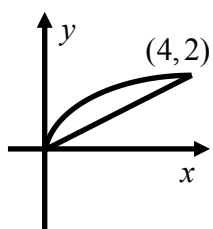
$$x: 0 \rightarrow 1$$

$$y: x^2 \rightarrow x$$

$$\begin{aligned} & \iint_D (x-1)d\sigma \\ &= \int_0^1 dx \int_{x^2}^x (x-1)dy \\ &= \int_0^1 (x-1)(x-x^2)dx \\ &= \int_0^1 (2x^2 - x^3 - x)dx = -\frac{1}{12} \end{aligned}$$

练习 5.2: 交换积分次序 $\int_0^2 dy \int_{y^2}^{2y} f(x,y)dx$

解:

换成 x 型

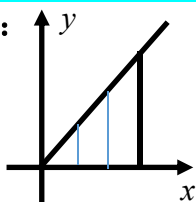
$$x: 0 \rightarrow 4$$

$$y: \frac{1}{2}x \rightarrow \sqrt{x}$$

$$\begin{aligned} & \int_0^2 dy \int_{y^2}^{2y} f(x,y)dx \\ &= \int_0^4 dx \int_{\frac{1}{2}x}^{\sqrt{x}} f(x,y)dy \end{aligned}$$

练习 5.3: 计算二重积分 $\int_0^1 dy \int_y^1 \frac{\sin x}{x} dx$

解:

换成 x 型

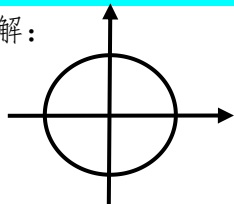
$$x: 0 \rightarrow 1$$

$$y: 0 \rightarrow x$$

$$\begin{aligned} \int_0^1 dy \int_y^1 \frac{\sin x}{x} dx &= \int_0^1 dx \int_0^x \frac{\sin x}{x} dy = \int_0^1 \left[\frac{\sin x}{x} y \right]_0^x dx \\ &= \int_0^1 \sin x dx = 1 - \cos 1 \end{aligned}$$

练习 5.4: 求 $\iint_D e^{x^2+y^2} dxdy$ D 为 $x^2+y^2 \leq a^2$ 围成。

解:



$$\theta: 0 \rightarrow 2\pi$$

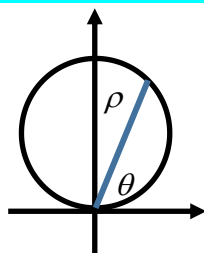
$$\rho: 0 \rightarrow a$$

$$\begin{aligned} & \iint_D e^{x^2+y^2} dxdy \\ &= \int_0^{2\pi} d\theta \int_0^a e^{\rho^2} \rho d\rho \\ &= \frac{1}{2} \int_0^{2\pi} d\theta \int_0^a e^{\rho^2} d\rho^2 \\ &= \frac{1}{2} \int_0^{2\pi} \left[e^{\rho^2} \right]_0^a d\theta \\ &= \pi(e^{a^2} - 1) \end{aligned}$$



练习 5.5: 求 $\iint_D \sqrt{x^2 + y^2} dx dy$ D 为 $x^2 + (y-1)^2 = 1$ 围成的区域

解:



$$\theta: 0 \rightarrow \pi$$

$$\rho: 0 \rightarrow 2 \sin \theta$$

$$\begin{aligned} & \iint_D \sqrt{x^2 + y^2} dx dy \\ &= \int_0^\pi d\theta \int_0^{2\sin\theta} \rho \cdot \rho d\rho \\ &= \int_0^\pi \left[\frac{1}{3} \rho^3 \right]_0^{2\sin\theta} d\theta \\ &= \frac{8}{3} \int_0^\pi \sin^3 \theta d\theta \\ &= \frac{32}{9} \end{aligned}$$

